Information Sufficiency for Reserve-Risk Screening under Degraded Wind-Turbine SCADA Telemetry

Junyu Li and Juntao Du* School of Statistics and Applied Mathematics, Anhui University of Finance and Economics, Bengbu 233030, China
*Corresponding author: dujuntao@aufe.edu.cn

***Abstract*—Pre-dispatch operating reserve screening near the wind-turbine aerodynamic boundary is governed by asymmetric shortfall risk. We study how the minimum sufficient model changes when SCADA telemetry is delayed or blade pitch is unavailable, separating state recoverability from reserve-decision value. On the 134-turbine WTB benchmark, deterministic aerodynamic rules achieve the lowest evaluated boundary-band surrogate cost under fresh telemetry (589,535 kW · h at 10 min; 881,367 kW · h at 60 min). Under synthetic 10–30 min delays, non-neural state-conditional recalibration absorbs 36.3%–61.7% of within-condition shortage loss; absorption reaches 66.5% at the 60-min stress endpoint. With pitch withheld at deployment but available during historical training, active-power consequences retain boundary information (NMI = 0.461), while representation metrics are not interpreted as decision-value evidence. The latest imported strong-baseline snapshot gives posterior conditioning a +445,639 kW·h full-population penalty relative to site-local wind-speed quantiles; this candidate remains pending deep audit. The diagnostics motivate comparisons of physical rules, recalibration and direct local quantiles by information regime; they do not certify model sufficiency or universal neural superiority. External transfer evidence is limited to a deterministic bounded branch and does not establish universal cross-site value.**

***Index Terms*—Wind power reserve, SCADA telemetry degradation, latent operating boundary, state-conditional recalibration, minimum sufficient model, consequence-based recovery.**

I. Introduction

In bulk power system operations, real-time generation shortfalls incur severe asymmetric shortage penalties relative to surplus headroom; conventional symmetric metrics like root-mean-square error (RMSE) mask severe tail shortfall events that govern operating reserve adequacy in Level-1 pre-dispatch screening [1], [2], [3], [4], [5]. This operational vulnerability concentrates near the aerodynamic demarcation boundary between Maximum Power Point Tracking (MPPT, Region 2) and active blade-pitch regulation (Region 3). Near rated inflow velocity, local power sensitivity transitions abruptly from a steep cubic trajectory ($\partial P/\partial v \propto v^2$) toward zero as pitch actuators shed aerodynamic lift [6], [7]. When operating states shift across this threshold unannounced, deterministic power curves over-extrapolate generation into Region 3, precipitating severe unhedged reserve deficits.

This vulnerability is exacerbated by telemetry degradation in commercial Supervisory Control and Data Acquisition (SCADA) systems, including gateway serialization overhead, packet queuing backlogs, and intermittent cellular transmission jitter [8], [9]. In operational archives (such as Kelmarsh), 99.6% of discrete turbine status transitions occur between standard 10-minute polling boundaries. Furthermore, in third-party aggregator and Virtual Power Plant (VPP) triage, blade-pitch availability is treated here as an imposed evaluation condition; its commercial prevalence is not established. When delayed or missing measurements enter physical power equations, dispatch systems clear reserve margins against obsolete operating points. With pitch withheld, the control register is unavailable, but the evaluated valid regime labels remain separable by fresh wind speed; consequence-signal classification therefore tests redundant information rather than necessary latent-state recovery ($Z_t \to P_t, Q_t$, trajectory).

Existing literature frequently treats aerodynamic curves and deep neural networks as competing predictive paradigms [10], [11], [12], [13], [14], [15], [16], overlooking their distinct operational boundaries under information freshness constraints. Rather than asking which model achieves marginal predictive gains, bulk power system operations require answering a foundational engineering question:

$$\boxed{\begin{array}{c}\text{What is the minimum sufficient model}\\ \text{under changing telemetry observability?}\end{array}} \tag{1}$$

To resolve this foundational question, we evaluate reserve screening across a three-stage inferential ladder: Observability → Recoverability → Decision Sufficiency. The 134-turbine WTB benchmark serves as the primary mechanism environment (five seeds 201–205), complemented by multi-year evaluations on three European facilities (ENGIE La Haute Borne, Kelmarsh, Penmanshiel) to delineate site-dependent boundary conditions without claiming a universal transfer law; turbine count alone does not explain the observed cross-site variation. Treating spatio-temporal graph and mixture-of-experts (MoE) formulations strictly as implementation comparators rather than novel modeling contributions (confirming MoE functions strictly as an implementation comparator, $p = 0.380$), this investigation establishes three substantive contributions across SCADA observability regimes:

1. **Regime 1 — Aerodynamic Sufficiency and Reliability Breakdown**: In boundary-active operating regimes ($|v_{\text{phys}} - 10.5| \le 1.0$ m/s, $N = 13{,}883$ cell events), deterministic aerodynamic quantile rules have the lowest evaluated surrogate cost under fresh observable telemetry (589,535 kW · h at $h = 1$, 881,367 kW · h at $h = 6$, with observed ~ 6.8% violation), saving 8.1% (−77,659 kW · h) over neural architectures. Operational latencies of 10–30 min

expose the onset of cubic extrapolation errors, while a 60-min contingency lag demonstrates asymptotic reliability breakdown (24.0% ± 1.7% violation, 127.6 MWh deficit).

2. **Regime 2 — Sufficiency of Non-Neural Recalibration under Observable Drift**: We decouple telemetry staleness from genuine channel missingness, demonstrating that under full channel observability, non-neural state-conditional recalibration absorbs 36.3%–61.7% of within-condition shortage loss across the 10–30 min operational latency grid and 66.5% at the 60-min contingency endpoint (127.6 → 42.8 MWh), without parameter updates. Operational delay regime conclusions remain provisional pending full field packet latency validation (Exp-A).
3. **Regime 3 — Consequence Information and Decision Boundary**: Under pitch-withheld deployment, active-power consequences retain boundary information (NMI = 0.461) in the privileged-training protocol, while fresh wind speed already separates the evaluated valid regime labels. The latest imported strong-baseline snapshot reports a full-population posterior penalty of +445,639 kW · h relative to site-local wind-speed bins; this numeric candidate remains pending deep audit. Its transition-versus-steady interaction interval crosses zero, so transition-localized heterogeneity is not currently established. The external deterministic branch is bounded evidence only and does not close EXP-01.

II. Related Work

A. From Average Wind-Power Accuracy to Transition-Window Risk

Spatio-temporal graph neural networks (STGNNs) advance multi-step wind plant forecasting by modeling spatial cross-correlations and wake advection [10], [11], [12], [13], with physics-informed extensions incorporating wake and boundary constraints [14], [15], [16]. While these architectures reduce plant-wide L_2 error, continuous trajectory minimization obscures discrete turbine operating state transitions between MPPT (Region 2) and pitch regulation (Region 3) that govern spinning reserve adequacy [17], [18], [19].

Recent physics-aware dynamic-graph mixture-of-experts (MoE) architectures, such as that of Zhao et al. [20], test whether coupling spatial learning, physical guidance, and expert routing improves point forecasting accuracy. Rather than iterating on forecasting network topologies, our formulation isolates the upstream information-and-decision problem: when critical aerodynamic states become unobservable due to telemetry degradation, what regime information remains recoverable from secondary consequence channels, when non-neural conditional recalibration suffices without parameter updates, and whether the recovered latent state alters asymmetric reserve-shortfall decisions. Probabilistic forecasts translate predictive uncertainty into reserve allocation [1], [2], [21]. We address this operational triage problem: screening aerodynamic transition risks under degraded SCADA telemetry prior to market clearing, functioning strictly as a Level-1 pre-dispatch proxy while abstracting downstream transmission constraints.

III. Problem Formulation and Latent Operating-Boundary Framework

A. Level-1 Pre-Dispatch Risk Screening and Asymmetric Decision Problem (PSREI)

Operating reserve screening functions as a pre-dispatch risk assessment mechanism to mitigate costly real-time generation imbalances [21], [22].

Hierarchical grid dispatch interface: Our formulation is strictly bounded to Level 1 pre-dispatch operating reserve screening proxies at the plant EMS or aggregator terminal, abstracting Level 2 bulk transmission AC-OPF and wholesale balancing settlements.

Penalized Reserve-Shortfall Energy Index (PSREI): Let $\hat{y}^{\text{fixed}}_{i,t}$ be the canonical scheduled point forecast, frozen before reserve-policy evaluation and shared across all compared policies, and let $y_{i,t}$ be realized generation. Shortfall is $s_{i,t} = \max(\hat{y}^{\text{fixed}}_{i,t} - y_{i,t}, 0)$. For upward reserve r_b and dimensionless penalty-to-procurement ratio $\rho \equiv c_u/c_r$ (balancing marginal unserved shortage penalty c_u against marginal reserve capacity cost c_r), the **Penalized Reserve-Shortfall Energy Index (PSREI)** in kW · h is:

$$C(r_b;\rho) = \sum_{(i,t)} \left[r_b(i,t) + \rho \max\left(\hat{y}^{\text{fixed}}_{i,t} - y_{i,t} - r_b(i,t),\, 0\right)\right] \Delta t. \tag{2}$$

Differentiating $\mathbb{E}[C(r_b;\rho)]$ yields the first-order condition $\partial \mathbb{E}[C]/\partial r_b = 1 - \rho \Pr[s_{i,t} > r_b] = 0$, giving the Newsvendor critical fractile $q^*(\rho) = 1 - 1/\rho$ [1]. For $\rho = 10$, $q^* = 0.90$, establishing a nominal 10% violation target.

Loss geometry and pinball quantile head: Point forecasting minimizes symmetric L_2 loss converging to conditional mean $\mathbb{E}[y \mid \mathbf{x}]$, which departs from upper quantiles in asymmetric operations ($\rho = 10$). Upward reserve margin $\hat{r}_{i,t}$ covering the frozen shortfall target $s_{i,t} = \max(\hat{y}^{\text{fixed}}_{i,t} - y_{i,t}, 0)$ is estimated via pinball loss:

$$\mathscr{L}_q(s,\hat{r}) = (s-\hat{r})\,(q - \mathbf{1}[s<\hat{r}]) = \begin{cases} q(s-\hat{r}), & s \geq \hat{r}, \\ (1-q)(\hat{r}-s), & s < \hat{r}. \end{cases} \tag{3}$$

Setting $q = q^* = 0.90$ aligns training with the Newsvendor condition: Bayes-risk-minimizing prediction converges to conditional shortfall quantile $\hat{r}^* = Q_{0.90}(s \mid \mathbf{x})$ [1], [2].

PCC portfolio smoothing & dispatch horizons: Net power at the Point of Common Coupling (PCC) bus aggregates turbine outputs ($P_{\text{farm},t} = \sum_{i=1}^{M} P_{i,t}$, $R_{\text{farm},t} = \sum_{i=1}^{M} R_{i,t}$), where turbulence cancels across the array, leaving boundary transitions as dominant unhedged risk. We evaluate immediate dispatch ($h = 1$, 10-min ahead) dominated by serial autocorrelation, and operational dispatch ($h = 6$, 1-hour ahead) where spatial wake dynamics govern decisions.

B. Latent Operating Regimes: MPPT, Pitch Regulation, and Demarcation Boundaries

Let $G = (V, E)$ denote a turbine array spatial graph with N nodes. For each turbine $i \in V$ at time step t, the system observes feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predicts active power $y_{i,t} \in \mathbb{R}$. Given observation history H and horizon P, the forecast mapping is:

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathscr{F}(\mathbf{X}_{t-H+1:t}, \mathscr{A}_{t-H+1:t}), \tag{4}$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H\times N\times F}$ and $\mathscr{A}_{t-H+1:t}$ denotes the graph adjacency sequence. We enforce strict causal boundaries: all input features, graph adjacency structures, and supervisory anchors depend solely on telemetry available at or before dispatch anchor t, with target trajectories commencing strictly at $t+1$. In WTB, instantaneous active power $\texttt{Patv}_{i,t}$ serves solely as an anchor-time operational status indicator, preventing look-ahead leakage.

At dispatch time t, turbine i operates in a discrete aerodynamic control regime governed by inflow wind speed $w_{i,t} = \texttt{Wspd}_{i,t}$ and collective blade pitch angle $\bar{p}_{i,t} = \frac{1}{3}\sum_{k=1}^{3} p_{i,t}^{(k)}$:

$$Z_{i,t} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad \text{(idle)}, \\ 1, & u_{\text{idle}} \le w_{i,t} \le u_{\text{rated}},\ \bar{p}_{i,t} < p_{\text{th}} \quad \text{(MPPT)}, \\ 2, & w_{i,t} > u_{\text{rated}},\ \bar{p}_{i,t} \ge p_{\text{th}} \quad \text{(pitch-regulated)}, \\ 3, & \text{otherwise} \quad \text{(transitional/ambiguous)}. \end{cases} \tag{5}$$

The evaluated label construction uses the fixed thresholds stored in the canonical cache metadata: $u_{\text{idle}} = 3.0$ m/s, $u_{\text{rated}} = 10.5$ m/s and $p_{\text{th}} = 2.0^\circ$ for WTB and the inspected external caches. These are operational label definitions in this experiment, not independently verified manufacturer-specific thresholds. Supervised alignment excludes ambiguous class $Z = 3$.

Under pristine telemetry, $\bar{p}_{i,t}$ and $w_{i,t}$ are known, so $Z_{i,t}$ is directly observable. Under transmission latency $\tau > 0$ or when blade-pitch registers are withheld across aggregator interfaces, the control register itself ceases to be accessible at deployment, while its dynamical electromechanical consequences ($Z_t \to P_t, Q_t,$ trajectory) remain partially preserved in arrived SCADA feeds. However, withholding pitch does not make the evaluated discrete labels unidentifiable: conditional on the declared validity mask, fresh wind-speed intervals separate all three valid classes. Ambiguous excluded cells are outside this result. The representation is consequently a supervised surrogate, not evidence of discovering a hidden physical state. The central inferential object is the soft operating-boundary posterior:

$$p(Z_{i,t} \mid \mathbf{X}_{\le t}), \tag{6}$$

inferred from historical trajectory, active-power transients, and spatial wake consequence channels.

C. Aerodynamic Power Conversion and the Stale-State Control Boundary

Mechanical power captured from inflow air is governed by $P_{\text{mech}}(t) = \frac{1}{2}\rho_{\text{air}}\pi R^2 C_p(\lambda(t), \beta(t))v^3(t)$, where $\rho_{\text{air}} \approx 1.225$ kg/m^3, R is rotor radius, $v(t)$ is wind speed, $\beta(t)$ is pitch angle, and $\lambda(t) = \omega_r(t)R/v(t)$ is tip-speed ratio [6], [7]. In Region 2 (MPPT, $u_{\text{idle}} \le v \le u_{\text{rated}}$), blades maintain nominal fine pitch $\beta \approx 0^\circ$ yielding cubic power growth $P(v) \propto v^3$; in Region 3 (Pitch regulation, $u_{\text{rated}} < v \le u_{\text{cut-out}}$), the pitch controller rotates blades to shed lift, clamping electrical output at nameplate capacity P_{rated}:

$$\left.\frac{\partial P}{\partial v}\right|_{v\to u_{\text{rated}}^-} = \frac{3}{2}\rho_{\text{air}}\pi R^2 C_{p,\max} u_{\text{rated}}^2 \gg 0, \qquad \left.\frac{\partial P}{\partial v}\right|_{v\to u_{\text{rated}}^+} \approx 0. \tag{7}$$

When SCADA telemetry experiences latency $\tau > 0$, aerodynamic curves evaluate stale tuples $(v_{t-\tau}, \beta_{t-\tau})$. If the turbine enters Region 3 during the latency window ($Z_{t-\tau} = 1$ while $Z_t = 2$), the delayed pitch register remains at fine pitch $\beta_{t-\tau} \approx 0^\circ$. Consequently, the aerodynamic rule extrapolates power along the cubic trajectory:

$$\hat{P}_{\text{phys}}(t) = \frac{1}{2}\rho_{\text{air}}\pi R^2 C_p(\lambda(t), 0^\circ)v^3(t) \gg P_{\text{rated}}. \tag{8}$$

Because output clamps at P_{rated}, this stale evaluation produces severe over-forecast error $\hat{P}_{\text{phys}}(t) - P_{\text{rated}} \gg 0$, surging empirical violation rates to 24.0% ± 1.7% under Delay-6 ($\rho = 10$), which acts as an asymptotic reliability breakdown stress endpoint.

D. The Non-Neural Baseline: State-Conditional Quantile Recalibration

To evaluate whether latency-induced drift can be absorbed by non-neural statistical correction, we formalize state-conditional quantile recalibration. Given delayed telemetry $\mathbf{s}_{i,t-\tau} = (w_{i,t-\tau}, \bar{p}_{i,t-\tau})$, we partition the state space into K bins $\{\mathscr{B}_k\}_{k=1}^K$ defined by cut-in, rated wind speed, and pitch thresholds ($Z_{i,t-\tau} = k$). To eliminate data leakage, empirical quantile reserve margins $\hat{r}(k)$ are calibrated strictly on held-out validation residuals $\mathscr{D}_k^{\text{val}}$:

$$\hat{r}(k) = \inf\left\{ r \in \mathbb{R} : \frac{1}{|\mathscr{D}_k^{\text{val}}|} \sum_{(i,t)\in\mathscr{D}_k^{\text{val}}} \mathbf{1}\left[\hat{y}_{i,t}^{\text{phys}} - y_{i,t} \le r\right] \ge q^* \right\}, \tag{9}$$

where $\mathscr{D}_k^{\text{val}} = \{(i,t) \in \mathscr{D}_{\text{val}} : \mathbf{s}_{i,t-\tau} \in \mathscr{B}_k\}$ and $q^* = 1 - 1/\rho = 0.90$.

We evaluate two calibration protocols: (1) *Clean-Frozen Calibration*, where margins $\hat{r}^{\text{clean}}(k)$ are calibrated on clean validation data ($\tau = 0$) and strictly frozen; and (2) *Condition-Matched Calibration*, where for a known delay τ, margins $\hat{r}(k;\tau)$ are re-estimated on held-out validation residuals with matching delay τ. Recalibration is an **information-preserving correction**: it updates residual margins conditional on observed bins, but cannot synthesize missing state information when blade pitch is withheld.

E. Latent State Recovery from Spatio-Temporal Consequence Channels

Physical dynamic wake graph & encoder: For WTB, time-varying directed wake graphs $\mathscr{A}_t$ link turbine pairs within aerodynamic wake cones ($\alpha = 25^\circ$) weighted by streamwise and cross-stream decay [14], [16]. A directed-diffusion GRU encoder within a 110,012-parameter forecasting model aggregates messages via:

$$\begin{aligned} \mathbf{X}_t^{(\ell+1)} = \text{LN}\Big(&\mathbf{W}_{\text{self}}^{(\ell)}\mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)}\text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathscr{A}_t) \\ &+ \mathbf{W}_{\text{out}}^{(\ell)}\text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathscr{A}_t) + \mathbf{R}^{(\ell)}\mathbf{X}_t^{(\ell)}\Big), \end{aligned} \tag{10}$$

and extracts node-level representations $\mathbf{h}_{i,t}$. When blade pitch $\bar{p}_{i,t}$ is withheld, the encoder estimates the declared regime posterior $p(Z_{i,t} \mid \mathbf{X}_{\le t})$ from secondary consequence channels: active power transients ($\texttt{Patv}_{i,t}$), electrical responses, and dynamic wake advection, strictly excluding $\texttt{Pab_mean}_{i,t}$.

The primary wind-speed-versus-posterior comparison uses the 87,067-parameter dense-classifier variant, whereas external transfer uses the 110,012-parameter MoE variant. The unrouted dense forecasting control has115,240 parameters and is approximately, not exactly, capacity matched. The channel ablation is a separate MLP probe using latest, mean, standard

deviation and maximum history features. It is not an intervention on the full spatio-temporal encoder. The strong-baseline decision comparison freezes the point forecast, calibrates five posterior-probability bins or ten wind-speed bins on validation shortfalls, and evaluates their 0.90 quantiles on identical test targets. Those comparisons must not be described as one end-to-end quantile network.

F. Reserve-Tail Estimation and Residual Quantile Sizing

To study balancing exposure, reserve margins target the asymmetric shortfall distribution rather than symmetric point forecasts, separating: (1) posterior operating-state inference $\pi_{i,t}^{\text{pitch}} = p(Z_{i,t} = 2 \mid \mathbf{X}_{\le t})$; (2) scheduled point forecasting $\hat{\mathbf{y}}_{i,t+1:t+P}$; and (3) reserve-tail quantile sizing $\hat{r}_{i,t} = Q_{0.90}(s_{i,t} \mid \mathbf{X}_{\le t})$.

Directly estimating extreme quantiles on raw power trajectories via uncalibrated neural pinball regression introduces severe tail conservatism (>1.6M kW reserve, <1.2% violation, Table I). To resolve this pathology, estimation is decoupled: point trajectories $\hat{\mathbf{y}}$ anchor nominal expectations, while upward reserve margins $\hat{r}_{i,t}$ are estimated conditionally on forecast residuals:

$$\begin{aligned} \hat{r}_{i,t} &= \text{MLP}_{\text{res}}([\mathbf{h}_{i,t}; \mathbf{x}_{\text{anchor}}]), \\ &\text{minimized under } \mathscr{L}_{q^*}(s_{i,t}, \hat{r}_{i,t}), \end{aligned} \tag{11}$$

or binned empirically over held-out validation residuals conditioned on latent posterior $\pi_{i,t}^{\text{pitch}}$.

G. Conditional Minimum-Sufficient Decision Rule

The phrase "minimum sufficient'' is used here as a conditional, risk-constrained model-selection criterion rather than a universal ranking law. Let $\mathscr{M}(\mathscr{T}_t^{(d)})$ be the admissible reserve policies measurable with respect to the arrived information set, and let $b_{\mathscr{T}}$ be the strongest pre-registered direct conditional baseline under that same information set. With the canonical frozen point forecast $\hat{y}^{\text{fixed}}$ and common shortfall target $s_t = \max(\hat{y}_t^{\text{fixed}} - y_t, 0)$, define $C_P(m)$ as fixed-target PSREI and $V_P(m)$ as the violation rate on population P. For pre-registered α, cost tolerance ϵ_C, violation tolerance ϵ_V, absolute violation cap $v_{\max}$, and complexity preorder Complexity(m), select

$$\begin{aligned} m^* = \arg\min_{m \in \mathscr{M}(\mathscr{T})} &\text{Complexity}(m) \\ \text{subject to} \\ U_{1-\alpha}[C_P(m) - C_P(b_{\mathscr{T}})] &\le \epsilon_C, \\ U_{1-\alpha}[V_P(m) - V_P(b_{\mathscr{T}})] &\le \epsilon_V, \\ U_{1-\alpha}[V_P(m)] &\le v_{\max}. \end{aligned} \tag{12}$$

Here $U_{1-\alpha}$ is the one-sided UCB under the registered seed/block unit. E04 froze $\alpha = 0.05$, $\epsilon_C = \epsilon_V = 0$, $v_{\max} = 0.10$, and $q^* = 0.90$. On WTB Full, all four B/C/D/D10 policies exceeded the absolute violation UCB cap and the selector returned "no feasible policy.'' B/C/D/D10 denote wind-speed bins, posterior quantiles and two hybrid policies, not an architecture ranking. No model is certified minimum-sufficient or safe for deployment; tolerances remain unchanged. The conditional rule is falsifiable, not a universal law. E07 remains inactive.

H. Architecture Variants as Mechanism Ablations

To test whether dynamic Mixture-of-Experts (MoE) routing provides operational value over unrouted representations, we formulate three matched-capacity variants: 1. **STGQ-Modular** (Decoupled Canonical): Pairs frozen spatio-temporal representations $\mathbf{h}_{i,t}$ with a dedicated residual quantile head; 2. **STGQ-Dense** (Unrouted Joint): Couples spatial embeddings $\mathbf{h}_{i,t}$ directly to a shared linear projection for point forecasts $\hat{\mathbf{y}}$ and a joint dense quantile head; 3. **STGQ-Routed** (Dynamic MoE): Evaluates dynamic MoE routing [23], [24] combining expert predictions $\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^{E} g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t})$.

Training minimizes:

$$\begin{aligned} \mathscr{L} = \mathscr{L}_{\text{pred}} &+ \lambda_{\text{align}}\mathscr{L}_{\text{align}} + \lambda_{\text{aux}}\mathscr{L}_{\text{aux}} \\ &+ \lambda_{\text{smooth}}\mathscr{L}_{\text{smooth}} + \lambda_{\text{bal}}\mathscr{L}_{\text{bal}} + \lambda_{\text{force}}\mathscr{L}_{\text{force}}, \end{aligned} \tag{13}$$

where $\mathscr{L}_{\text{align}}$ anchors logits to declared regimes $Z_{i,t}$ on unmasked samples ($M_{i,t} = 1$); $\mathscr{L}_{\text{force}}$ sharpens boundary discrimination; $\mathscr{L}_{\text{bal}}$ balances expert utilization; and $\mathscr{L}_{\text{aux}}, \mathscr{L}_{\text{smooth}}$ enforce spatial wake consistency.

Operating-Regime Representation Prior: The auxiliary loss $\mathscr{L}_{\text{align}}$ acts as an operating-regime prior embedded into representation space. The neural backpropagation pipeline contains no quantile loss, pinball penalty, or Newsvendor decision loss. Checkpoint selection is strictly governed by validation RMSE. Quantile reserve margins are calibrated on held-out validation residuals or estimated by the decoupled residual head, yielding a jointly represented boundary posterior rather than an end-to-end decision posterior.

IV. Case Study Configuration and Operational Constraints

A. Datasets and Preprocessing

Empirical evaluation spans the 134-turbine WTB benchmark and the ERA5 thermodynamic contrast patch. WTB (KDD Cup 2022) comprises 134 turbines over 245 days of 10-minute SCADA telemetry (T = 35,280, N = 134, F = 11) [25], with chronological 180/30/35-day train/val/test splits ($H = 36, P = 24$). ERA5 covers three archived months on a 16×16 grid (T = 2208, N = 256, 8 features) [26] under 1325/441/442 splits. Multi-year SCADA records from three European commercial wind plants (ENGIE La Haute Borne, Kelmarsh, Penmanshiel) provide diverse turbine architectures for cross-farm evaluation.

B. Controlled Telemetry Degradation Protocols and Realism Mapping

To evaluate resilience under communication stress, we simulate three degradation tiers across five seeds (201–205): (1) **Tier 1 (Nominal Polling, 0–10 min, Clean)**: fresh SCADA on 10-min grid (IEC 61400-25); (2) **Tier 2 (Plausible Degraded Telemetry, 10–30 min, Delay-1 to 3)**: transmission lags from substation buffer backlogs [9], characterizing the operational transition regime; and (3) **Tier 3 (Contingency Stress Envelope, 60 min, Delay-6)**: a 6-step synthetic lag evaluating asymptotic physical breakdown [27], [28] as a stress endpoint rather than an industrial prevalence claim. Perturbations include additive Gaussian noise (σ_v = 1.0 m/s, σ_β = 2.0°) and a two-state Markov burst dropout chain ($p_{GB} = 0.08, p_{BB} = 0.75, d \le 6$) [9].

Fig. 1 separates information regimes and their unresolved decision tests; it is not an operational dispatch gate.

Arrived information $\mathcal{I}_t$

Fresh wind and pitch → evaluate physical rules.

Observable delayed channels → evaluate historical recalibration.

Pitch withheld → test consequence information against direct local quantiles.

For every candidate: matched frozen targets, paired uncertainty and the absolute risk cap.

E04: no feasible policy. E07: inactive.

Fig. 1. Conceptual information and evidence checks. Candidate families are hypotheses for comparison; the diagram neither certifies a deployment route nor converts regime recoverability into decision value.

C. Pre-Dispatch Risk Screening Framework and Resampling Hierarchy

The declared backbone recipe uses AdamW, validation-RMSE stopping, clipping and training mixed precision across seeds 201–205; auxiliary probes and quantile heads have distinct budgets. This recipe does not bind every historical execution environment or establish evaluation precision. Validation calibration precedes the declared test replay; complete historical input and execution binding remains unverified. Uncertainty is experiment-specific: earlier diagnostics report 1,000 paired block-bootstrap draws, whereas E04/E05 follow their own registered resampling. The 35 native test blocks start from the first issue anchor at 144-step intervals, not calendar-day boundaries. Five seeds measure initialization variation, not independent farms or test periods.

V. Empirical Findings: Minimum Sufficient Modeling Across Observability Regimes

A. Regime 1: Fresh Observable Telemetry → Deterministic Aerodynamics and Operational Breakdown Boundaries (RQ1)

Table I summarizes the multi-day replay dispatch benchmark for 1-hour ahead operational dispatch ($h = 6$) evaluated on the boundary-active operating slice ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13{,}883$ cell observations across 134 turbines over the 35-day test split) under nominal clean telemetry. Table II reports seed-paired hypothesis tests and mechanism ablations across operational regimes, along with 95% confidence intervals from a paired daily cluster bootstrap over the 35 observed test days (1,000 resamples).

The empirical evidence directly establishes the operational sufficiency of deterministic aerodynamics under fresh observable telemetry. Under pristine telemetry ($\tau = 0$), Continuous Physical Quantile achieves the lowest surrogate reserve screening cost within the active boundary band ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13{,}883$ cell observations) among all evaluated methods (589,535 $\pm$ 77,390 kW · h at immediate dispatch $h = 1$ and 881,367 $\pm$ 95,741 kW · h at 1-hour dispatch $h = 6$, with observed $\sim$ 6.8% violation). It outperforms both unrouted dense networks (STGQ-Dense: 991,181 kW · h) and dynamically routed architectures (STGQ-Routed: 959,027 kW · h) by 8.1%–11.1%. As confirmed in Table II, Continuous Physical Quantile achieves a −77,659 kW · h advantage over neural representations, with a 95% daily cluster bootstrap confidence interval (1,000 paired resamples over 35 observed days) strictly excluding zero ([−106,175, −49,143]). When inflow wind speed and blade pitch angle are fresh and directly observable, deterministic aerodynamic equations yield the lowest evaluated surrogate reserve screening cost. Neural networks, lacking hard physical saturation bounds, introduce unnecessary variance under nominal telemetry.

Table I exposes the operational reliability boundary of physical rules when telemetry becomes stale. Transmission delays of 10–30 minutes introduce latency drift between true inflow and evaluated registers. Under a 60-minute latency contingency stress test (Delay-6), which serves as an asymptotic reliability breakdown stress endpoint rather than an industrial prevalence claim, physical equations evaluate stale state tuples $(v_{t-\tau}, \beta_{t-\tau})$. Near rated wind speed, the localized slope discontinuity (7) amplifies delayed-inflow errors: turbines that have crossed into active blade-pitch regulation ($Z_t = 2$) are evaluated with obsolete fine pitch ($\beta_{t-\tau} \approx 0°$), over-extrapolating power output along the steep cubic trajectory (8). Unadapted physical quantile violation rates surge from 6.81% to 24.02% $\pm$ 1.69% at $h = 1$ and 12.20% $\pm$ 1.25% at $h = 6$ (1,136,466 kW · h), breaching the nominal 10% Newsvendor target ($q^* = 0.90$) and leaving 127.55 MWh of unhedged generation shortfall across the array.

B. Regime 2: Stale but Observable Telemetry → Non-Neural Recalibration Absorbs Drift (RQ2)

When telemetry channels remain observable under transmission delays, condition-matched state-conditional recalibration (9) adjusts quantile margins using held-out validation residuals undergoing the identical delay. As shown in Table III, state-conditional recalibration absorbs 36.3% of within-condition shortage loss at $\tau = 10$ min (37.7 → 24.0 MWh), 53.9% at $\tau = 20$ min (51.8 → 23.9 MWh), and 61.7% at $\tau = 30$ min (69.1 → 26.4 MWh), holding fleet violation at 7.1%–7.5% and a surrogate reserve cost of 1,228,609 kW·h at the $h = 1$ contingency endpoint. Under full channel observability, recalibrated physical curves outperform all neural models (STGQ-Routed: 1,274,047 kW · h; STGQ-Modular: 1,283,855 kW · h).

A mismatched calibration matrix across test latencies $\tau_{\text{test}} \in [0, 60]$ min and calibration horizons $\tau_{\text{cal}} \in [0, 60]$ min establishes the operational limits of recalibration under communication drift. Recalibrating at $\tau_{\text{cal}} = 10$ min withstands latency drift up to $\tau_{\text{test}} = 20$ min while maintaining target-satisfying tail coverage (8.2%–9.1% violation), but breaks down when staleness surges to $\tau_{\text{test}} = 60$ min (13.8% violation). Under full channel observability, distribution drift does not require neural representation learning: state-conditional recalibration recovers 36.3%–66.5% of the observed shortage loss across the evaluated latency grid (127.6 → 42.8 MWh at the 60-min contingency endpoint, a 66.5% absorption at 9.5% violation) without updating network weights.

The historical delay scan uses condition-specific forecasts and an arrival-wind-selected boundary population. Its clean-versus-recalibrated physics comparison shares targets and masks within a condition, but comparisons across delays mix forecast and population changes. Older dense/MoE phase-scan costs also used different shortfall targets. Those fixed-target/fixed-population interpretations are withdrawn. The corrected replay freezes canonical shortfalls and physical-time boundary cells across methods and conditions, and retains a ten-bin wind-

TABLE I

CROSS-SEED MULTI-REGIME PRE-DISPATCH RESERVE SCREENING BENCHMARK ON BOUNDARY-ACTIVE OPERATING REGIMES UNDER CLEAN-CALIBRATED PROTOCOL (h = 6, 1-HOUR AHEAD DISPATCH, 5 SEEDS 201–205). MEAN ± STANDARD DEVIATION ACROSS BOUNDARY-BAND EVENTS ($|v_{\text{PHYS}} - 10.5| \leq 1.0$ M/S, N = 13,883 CELL OBSERVATIONS, 35 TEST DAYS, 134 TURBINES, Δt = 10 MIN, COST RATIO ρ = 10). ALL QUANTILE BINS AND MULTIPLIERS CALIBRATED EXCLUSIVELY ON NOMINAL CLEAN VALIDATION DATA. COMMENSURATE PLANT-WIDE FULL-POPULATION BASELINES (N = 408,939 CELLS) RANGE FROM 14.0M TO 16.5M KW · H (SEE TEXT AND METRIC-SCOPE REGISTRY).

Regime	Evaluated Model	Penalized Reserve Energy (kW·h)	Violation Rate	Reserve Energy (kW·h)	Shortage Energy (kW·h)	Pinball Loss
Clean (boundary)	Continuous Physical Quantile	**881,367 ± 95,741**	6.81% ± 1.13%	653,738 ± 72,557	22,763 ± 4,013	31.41 ± 2.66
	Direct Quantile MLP‡	1,649,721 ± 285,682	0.58% ± 0.26%	1,630,572 ± 293,346	1,915 ± 958	64.62 ± 12.05
	Global Quantile	998,988 ± 107,360	6.46% ± 1.31%	771,304 ± 95,661	22,768 ± 4,854	36.50 ± 3.41
	STGQ-Dense	991,181 ± 110,211	6.54% ± 1.04%	761,480 ± 94,244	22,970 ± 4,121	36.16 ± 3.58
	STGQ-Routed	959,027 ± 103,290	7.10% ± 1.17%	707,249 ± 91,305	25,178 ± 4,816	34.77 ± 3.16
	Missingness-Aware GBDT	1,238,603 ± 112,051	4.50% ± 0.73%	1,085,168 ± 118,015	15,343 ± 2870	46.85 ± 3.48
Delay-6	Continuous Physical Quantile	1,136,466 ± 84,101	12.20% ± 1.25%†	662,335 ± 73,483	47,413 ± 3,091	38.85 ± 2.06
	Direct Quantile MLP‡	1,682,251 ± 284,323	1.20% ± 0.52%	1,651,979 ± 297,715	3,027 ± 1,594	62.15 ± 11.70
	Global Quantile	1,180,603 ± 91,754	10.85% ± 1.55%†	780,916 ± 96,853	39,969 ± 6,087	40.73 ± 2.43
	STGQ-Dense	1,179,859 ± 95,268	11.04% ± 1.22%†	773,103 ± 94,462	40,676 ± 4,788	40.70 ± 2.66
	STGQ-Routed	1,163,593 ± 91,299	11.82% ± 1.64%†	720,015 ± 92,196	44,358 ± 7,341	40.01 ± 2.54
	Missingness-Aware GBDT	1,320,003 ± 93,135	8.36% ± 0.97%	1,030,858 ± 114,266	28,915 ± 4,469	46.68 ± 2.45
Noise	Continuous Physical Quantile	**998,273 ± 106,253**	7.59% ± 0.95%	746,697 ± 82,474	25,158 ± 3,968	33.97 ± 2.78
	Direct Quantile MLP‡	1,639,909 ± 242,922	0.80% ± 0.28%	1,618,439 ± 249,374	2,147 ± 953	60.62 ± 9.85
	Global Quantile	1,053,552 ± 114,150	7.48% ± 1.48%	802,361 ± 99,513	25,119 ± 5,822	36.26 ± 3.24
	STGQ-Dense	1,048,083 ± 115,181	7.63% ± 1.13%	792,472 ± 92,391	25,561 ± 4,643	36.04 ± 3.28
	STGQ-Routed	1,027,077 ± 114,317	7.62% ± 1.09%	769,864 ± 106,672	25,721 ± 3,963	35.16 ± 3.19
	Missingness-Aware GBDT	1,314,640 ± 129,158	4.66% ± 0.80%	1,160,329 ± 137,650	15,431 ± 3,099	47.11 ± 3.87
Markov	Continuous Physical Quantile	**907,871 ± 96,820**	7.20% ± 1.15%	663,875 ± 73,821	24,400 ± 4,106	31.59 ± 2.61
	Direct Quantile MLP‡	1,680,339 ± 289,602	0.57% ± 0.25%	1,660,135 ± 298,104	2,020 ± 992	64.38 ± 12.00
	Global Quantile	1,030,131 ± 108,592	6.87% ± 1.39%	785,305 ± 97,397	24,483 ± 5,104	36.78 ± 3.35
	STGQ-Dense	1,020,078 ± 112,195	6.90% ± 1.07%	775,899 ± 96,235	24,418 ± 4,246	36.35 ± 3.57
	STGQ-Routed	990,625 ± 104,487	7.51% ± 1.22%	720,858 ± 92,754	26,977 ± 5,183	35.10 ± 3.10
	Missingness-Aware GBDT	1,261,941 ± 112,754	4.88% ± 0.75%	1,094,730 ± 120,612	16,721 ± 3,298	46.62 ± 3.40

†Exceeds nominal 10% violation target (q^* = 0.90). Bold indicates lowest surrogate metric. **Scope & Reconciliation:** Evaluated at h = 6 strictly across boundary-band cells ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, N = 13,883, ~ 2.4% of test split); Continuous Physical Quantile achieves lowest cost (881,367 kW · h) but collapses to 12.20% violation under Delay-6. Full-population (N = 408,939) direct quantiles achieve 14.0M–16.5M kW · h at h = 6 and 6.49M–8.05M at h = 1 (Section V-D; incommensurate scopes). ‡Frozen-Embedding Direct Quantile MLP evaluates uncalibrated pinball regression on frozen representations; calibrated STGQ-Modular resolves this conservatism.

TABLE II

SEED-PAIRED OPERATIONAL HYPOTHESIS TESTS AND MECHANISM ABLATION CONTRASTS (h = 6, 1-HOUR AHEAD DISPATCH, COST RATIO ρ = 10).

Regime	Evaluated Contrast	ΔPSREI (kW·h)	95% Daily Cluster CI (1,000 Resamples)
Clean	Cont. Phys. (Rule)	−77,659	[−106.2k, −49.1k]**
	STGQ-Dense vs. Routed	+32,154	[−1.1k, +65.5k]
	Global Quantile	+39,962	[+12.6k, +67.4k]*
	Missingness GBDT	+279,576	[+249.5k, +309.6k]*
	Direct MLP (Pinball)	+690,695	[+448.8k, +932.6k]**
Delay-6	Cont. Phys. (Breakdown)	−27,126	[−46.9k, −7.3k]†
	STGQ-Dense vs. Routed	+16,267	[+7.1k, +25.4k]*
	Global Quantile	+17,010	[+3.5k, +30.5k]*
	Missingness GBDT	+156,411	[+140.0k, +172.8k]*
	Direct MLP (Pinball)	+518,658	[+307.2k, +730.1k]**
Noise	Cont. Phys. (Robust)	−28,804	[−47.6k, −10.1k]*
	STGQ-Dense vs. Routed	+21,006	[+2.4k, +39.6k]*
	Global Quantile	+26,475	[+9.1k, +43.8k]*
	Missingness GBDT	+287,563	[+248.8k, +326.3k]*
	Direct MLP (Pinball)	+612,832	[+413.7k, +811.9k]**
Markov	Cont. Phys. (Rule)	−82,754	[−111.0k, −54.5k]**
	STGQ-Dense vs. Routed	+29,453	[−2.9k, +61.8k]
	Global Quantile	+39,506	[+16.0k, +63.1k]*
	Missingness GBDT	+271,316	[+245.0k, +297.6k]*
	Direct MLP (Pinball)	+689,714	[+448.7k, +930.7k]**

Note: 95% CIs from paired daily cluster bootstrap over 35 observed test days (144 steps/cluster, 1,000 Monte Carlo resamples).
**$p < 0.01$,* $p < 0.05$. †Physical rule breaches nominal 10% violation target (12.20%). Dynamic MoE routing advantage over unrouted dense architecture is falsified (p = 0.380).

speed comparator.

C. Regime 3: Pitch Withheld: Information Sufficiency → Consequence-Based State Recovery and Decision Boundaries (RQ2)

The role of consequence-based state recovery is examined when blade-pitch telemetry is withheld across aggregator boundaries (`no_pitch`, Table III). Under pitch unobservability, physical rules and recalibration schemes lose direct state awareness, inflating recalibrated screening penalties from 824,393 kW · h at τ = 0 to 1,378,900 kW · h at τ = 60 min.

Under this regime, the control register is unobservable at deployment, while its electromechanical consequences ($Z_t \rightarrow P_t, Q_t$, trajectory) remain partially recorded in arrived SCADA feeds. Learned representations (STGQ-Modular) reconstruct operating states from secondary consequences under **offline privileged supervision**.

We provide an explicit disclosure that pitch is available in training inputs but withheld in validation/test deployment: blade-pitch registers (β) were present in historical training buffers (x_{hist} channels 7–8) and anchor physics, while pitch was strictly zero-masked and withheld during validation and testing. Trained using historically available pitch information that is withheld at deployment, offline privileged supervision improves latent-state organization and reduces representation collapse, while matched ablations do not establish an independent downstream reserve-cost gain (paired difference

TABLE III

MULTIDIMENSIONAL OPERATIONAL PROGRESSION ACROSS TELEMETRY LATENCY $\tau \in [0, 60]$ MIN AND PITCH OBSERVABILITY ON BOUNDARY-ACTIVE REGIMES ($|v_{\text{PHYS}} - 10.5| \leq 1.0$ M/S, $N = 13{,}883$ CELLS, $h = 1$, 5-SEED MEAN ON WTB; NOMINAL $\tau = 0$ BASELINE 593,258 KW · H ALIGNS WITHIN 0.6% OF THE PRIMARY FROZEN 589,535 KW · H BOUNDARY BENCHMARK). PENALIZED RESERVE ENERGY IN KW·H ($\rho = 10$).

τ (min)	Pitch	Clean Phys. Metric (Viol.)	Recal. Phys. Metric (Viol.)	STGQ-Mod. Metric (Viol.)	STGQ-Rout. Metric (Viol.)	Operational Regime
0	All	**593,258** (7.3%)	**593,258** (7.3%)	663,347 (6.1%)	663,915 (7.3%)	Regime I (Lowest Cost)
0	None	961,024 (3.2%)	824,393 (5.4%)	**755,794** (5.8%)	756,257 (6.4%)	Regime II (Repr. Adv.)
10	All	753,606 (11.2%†)	**742,211** (7.4%)	786,366 (6.7%)	789,094 (7.6%)	Regime I (Recal. Suff.)
10	None	1,008,743 (4.8%)	928,028 (6.2%)	**866,753** (6.1%)	874,156 (6.6%)	Regime II (Repr. Adv.)
20	All	896,198 (14.0%†)	**841,734** (7.1%)	875,838 (7.4%)	879,750 (8.0%)	Regime I (Recal. Suff.)
20	None	1,062,646 (6.1%)	1,010,530 (6.2%)	**947,869** (6.7%)	959,934 (7.1%)	Regime II (Repr. Adv.)
30	All	1,069,266 (16.6%†)	**939,190** (7.5%)	987,760 (7.1%)	990,091 (7.6%)	Regime I (Recal. Suff.)
30	None	1,138,774 (7.5%)	1,101,292 (6.7%)	**1,054,592** (6.4%)	1,064,238 (6.7%)	Regime II (Repr. Adv.)
60	All	1,656,285 (24.0%†)	**1,228,609** (9.5%)	1,283,855 (9.6%)	1,274,047 (9.6%)	Regime III (Tail Risk)
60	None	1,437,280 (11.8%†)	1,378,900 (9.0%)	**1,330,318** (8.7%)	1,331,840 (8.8%)	Regime II (Repr. Adv.)

†Exceeds nominal 10% violation target ($q^* = 0.90$). Bold indicates lowest target-satisfying surrogate screening metric. Under $\tau = 60$ min, target satisfaction is 3/5 seeds (60%) across models.

ΔPSREI $= -312\text{k} \pm 1{,}086\text{k}$ kW · h, 95% bootstrap CI $[-1.31\text{M}, +0.34\text{M}]$, crossing zero).

Empirical consequence-channel identifiability audits disentangle the recovery mechanism: contemporaneous active power is the dominant mediator (AUROC = 0.988), while lagged trajectories and non-power consequence channels retain partial additional state information, and thermal channels contribute negligible high-frequency boundary information for 10-minute dispatch.

As detailed in Table III, STGQ-Modular achieves 755,794 kW · h at $\tau = 0$ (an 8.3% saving of 68,599 kW · h over recalibrated physics) and maintains the reported reserve-screening comparison against condition-matched recalibrated physics as a historical diagnostic only; no transition diagnostic is used as current decision evidence.

Separation of Information Recoverability and Decision Value: State recoverability does not imply universal decision superiority. Operational validation separates three distinct claims: (i) *Operating-state discrimination* (AUROC > 0.94, NMI 0.461, ARI 0.647); (ii) *Probabilistic calibration of* $P(Z \mid \mathcal{T}_t^{(d)})$ (bulk ECE $\leq 3.46\%$, satisfying the 5% screening gate; tail MCE ≈ 0.75–0.85); and (iii) *Reserve-decision value*. Against a strong observable wind-speed-conditioned quantile baseline (10 bins, 13.78M kW · h plant-wide), posterior conditioning does not reduce plant-wide reserve screening costs (latest imported candidate $\Delta L = L_{\text{posterior}} - L_{\text{wspd}} = +445{,}639$ kW · h; pending deep audit). The latest candidate does not certify a separate steady-state superiority or transition-localized heterogeneity claim; the prior slice-specific estimates are historical pending re-audit. Thus, strong wind-speed-conditioned quantiles remain lower-cost plant-wide.

The latest candidate reports a transition-slice penalty of +71,204 kW · h relative to site-local wind-speed bins. The imported transition-versus-steady interaction is −303,232 kW · h with 95% interval $[-597{,}953, +19{,}013]$ kW · h, crossing zero. These values are pending deep audit and do not establish transition-localized decision superiority.

The historical matched-budget frontier is retained as a diagnostic artifact but is not used as current claim evidence while the imported strong-baseline candidate undergoes deep audit. Its prior transition-localized allocation interpretation is therefore treated as pending rather than decisive.

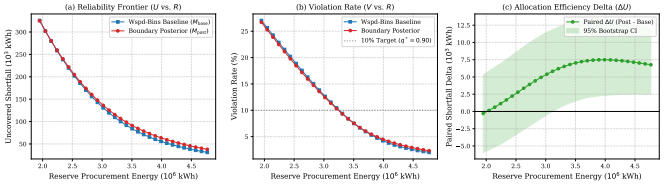


Fig. 2. Historical matched-budget frontier diagnostic in transition windows. The artifact is retained for provenance, but its prior allocation interpretation is not used as current claim evidence while the imported strong-baseline candidate undergoes deep audit.

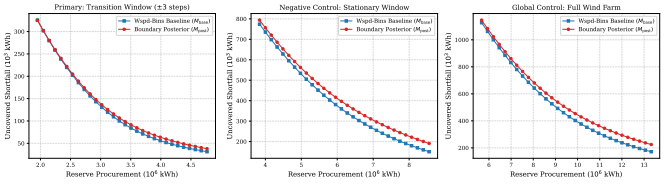


Fig. 3. Re-optimized penalty-ratio sensitivity ($\rho \in [2, 100]$). In transition windows, baseline dominance widens from +22.4k kWh ($\rho = 5$) to +998.2k kWh ($\rho = 100$). Plant-wide, baseline is preferred across the entire evaluated range. Posterior advantage at $\rho = 2$ (transition only) vanishes for all $\rho \geq 5$.

Dynamic Newsvendor re-optimization across $\rho \in [5, 100]$ (Fig. 3) exhibits no economic crossover, as high conditional tail variance forces severe reserve over-procurement (+1.10M kWh at $\rho = 100$). Recovering the latent boundary does not establish reserve-allocation superiority over strong observable wind-speed conditioning.

D. Supporting Evidence and Falsification of Alternative Explanations (RQ3)

Historical modular diagnostics report 9.7% violation and no detected advantage of heuristic abstention over uniform margin expansion ($p > 0.40$). These point metrics do not certify the risk cap, causal attribution to joint learning or routing equivalence under common frozen targets. Classification accuracy alone does not establish reserve decision value.

External decision-value conclusions from the retrospective EXP-01 v4 calculation are withdrawn because its forecasts

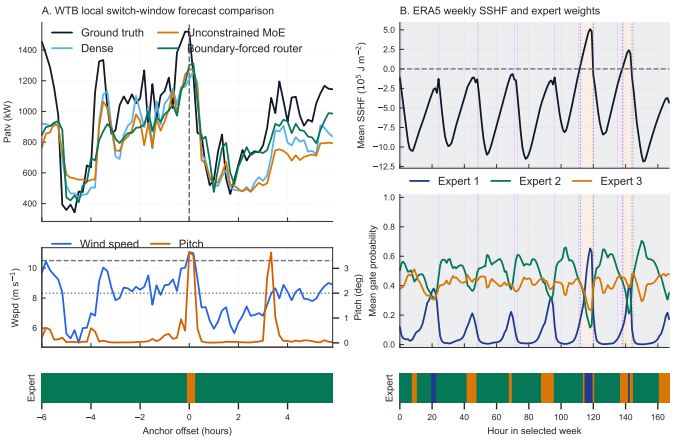


Fig. 4. Case-study view of the WTB local control-boundary switch window and the ERA5 weekly thermodynamic positive control. Panel A shows a WTB MPPT-to-pitch transition window (ground-truth active power, dense and boundary-forced router predictions, driving wind speed and pitch, and dominant-expert assignment); Panel B shows ERA5 mean surface sensible heat flux against weekly expert gate weights. This figure illustrates mechanism behavior on the primary site and does not compare cross-site reserve costs; the external-site comparisons are reported numerically in the text and in the supplementary site ledger.

and targets came from physical windows shifted by 120 days (Kelmarsh) or 720 days (Penmanshiel). The corrected-v2 benchmark is superseded and not reinstated as the current matched-decision result. The current bounded evidence is the deterministic_matched_transfer_v1 branch: at $h = 6$, the posterior-minus-local-wind-bin PSREI deltas are +6.3k kW · h for Kelmarsh (interval crossing zero) and +356.0k kW · h for Penmanshiel. This branch uses existing weights and local frozen forecasts/calibration, does not certify complete raw retraining, and does not close the full EXP-01 claim.

VI. Discussion: Minimum Sufficient Model Complexity

The evaluated diagnostics motivate the conditional criterion of **Minimum Sufficient Model Complexity**: when critical telemetry is unavailable, latent operating states may remain recoverable from observable consequence signals. Whether learned representations are necessary for that recovery, whether they change the risk shape, and whether the changed risk shape improves the downstream decision are three separate empirical questions. The evidential chain is therefore recoverability → risk-shape change → decision value, with no arrow assumed automatically. In particular, NMI/ARI/AUROC or calibration cannot substitute for matched PSREI evidence. Given the current strong local baseline, we do not add further graph, MoE, or hidden-state architectures; any complex representation must demonstrate deployment necessity under the pre-registered risk-constrained selector in E04. The following comparisons guide further assessment; none supplies deployment authorization:

1) Fresh and Observable → Physics: Deterministic aerodynamic curves evaluate pristine telemetry ($\tau = 0$) with asymptotic power conservation ($\partial P/\partial v \propto v^2$ below rated speed, $P \equiv P_{\text{rated}}$ above rated speed). Avoiding out-of-distribution neural variance, they achieve the lowest evaluated surrogate reserve screening cost within the boundary band ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13{,}883$ events: 589.5k–881.4k kW · h, observed ~ 6.8% violation; Tables~I and~III), without establishing a selector-certified minimum.

2) Stale but Observable → Recalibration: Under observable operational latency (10–30 min), state-conditional recalibration absorbs 36.3%–61.7% of within-condition shortage loss and maintains violation rates below the nominal 10% target (7.1%–7.5%) without updating neural weights. In the evaluated latency grid, static recalibration breaks down at the contingency horizon ($\tau = 60$ min, 13.8% violation), which acts as a synthetic reliability breakdown stress endpoint rather than an industrial prevalence claim.

3) Pitch Withheld: Information Sufficiency → Consequence Information: When blade-pitch telemetry is withheld (`no_pitch`), fresh wind speed still separates the evaluated valid regime labels. In the privileged-training deployment protocol, active-power consequences retain boundary information (NMI = 0.461). This representation result does not establish decision value; the historical cost contrasts against recalibrated physics are not used here as current decision evidence.

4) Operational Decision Boundary: The latest imported strong-baseline snapshot reports a full-population posterior penalty of +445,639 kW · h relative to site-local wind-speed bins. Its transition-slice penalty is +71,204 kW · h, and the imported transition-versus-steady interaction is −303,232 kW · h with 95% interval [−597,953, +19,013] kW·h, crossing zero. These values are pending deep numerical and statistical audit; they support no plant-wide posterior cost superiority and do not establish transition-localized heterogeneity. The historical closure and its slice-specific interaction are superseded provenance only. Because the current interaction interval crosses zero, no transition-localized heterogeneity or relative penalty attenuation is promoted to a current claim. A deployable hybrid policy is not certified to differ from wind-speed binning; the current candidate snapshot remains pending audit. MoE functions strictly as an implementation comparator rather than a required mechanism: increasing routing complexity is not the operative source of value under the tested observability regimes, and the older modular/dense metrics cannot establish routing equivalence under a common frozen target.

5) External Evidence Boundary: The current external evidence is limited to the deterministic_matched_transfer_v1 branch: at h=6, the bounded deltas are +6.3k kW · h for Kelmarsh (interval crossing zero) and +356.0k kW · h for Penmanshiel. This branch uses existing weights and local frozen forecasts/calibration and does not close the full EXP-01 claim. Corrected-v2 is superseded/not reinstated, while v4 is invalidated by physical-window mismatch; no universal cross-site transfer conclusion is claimed.

VII. Limitations

The empirical findings are bounded by four operational and experimental scope conditions:

1) Synthetic Stress: Impairments are evaluated under synthetic stress (10–30 min backlogs, 60-min latency, Gilbert-Elliott dropouts); field units may exhibit sensor icing or individual pitch actions outside supervisory records.

2) Risk Surrogacy: Evaluation operates at Level-1 pre-dispatch operating reserve screening via the PSREI newsvendor surrogate ($\rho = 10$), abstracting Level-2 bulk transmission AC-OPF and balancing settlements.

3) Horizon Scope: Benchmarks use a 35-day continuous test split across five random seeds (201–205), omitting multi-annual climatological cycles. Rated wind speeds vary by model (12.5 m/s on MM92 vs. 14.5 m/s on MM82), requiring OEM calibration.

4) Deployment Boundaries: EXP-01 v4 is invalidated by physical-window mismatch; its numerical cost differences and intervals are withdrawn from current conclusions. Historical EXP-02 also requires replay because its physical proxy labels used mismatched normalization statistics. These failures leave the corresponding decision questions unresolved. Historical pitch availability during training must remain distinguished from pitch-never-recorded deployment scenarios.

VIII. Conclusion

This study repositions machine learning in reserve screening from an indiscriminate forecasting tool to a targeted conditional inference mechanism governed by SCADA observability regimes. Telemetry degradation creates a hierarchy of sufficiency: *use the least complex model sufficient for the arriving information regime. When critical telemetry is unavailable, latent operating states may remain recoverable from observable consequence signals; whether learned representations are necessary for that recovery, and whether recovered state information improves downstream decisions, are separate empirical questions.* First, under fresh telemetry, deterministic aerodynamic power curves minimize surrogate reserve penalties within the boundary band ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13{,}883$ events: 589,535 kW · h at $h = 1$, 881,367 kW · h at $h = 6$, ~ 6.8% violation), collapsing only under unadapted latency (24.0% violation). Second, under observable delays, state-conditional recalibration absorbs 36.3%–61.7% of within-condition shortage loss across the 10–30 min operational grid, rising to 66.5% at the 60-min contingency endpoint (127.6 → 42.8 MWh), without updating neural weights. Third, when blade pitch is withheld, active-power consequences retain boundary information ($NMI = 0.461$) under the privileged-training protocol. The latest imported strong-baseline snapshot reports a full-population posterior penalty of +445,639 kW·h relative to site-local wind-speed bins, but this candidate remains pending deep audit. Its transition-versus-steady interaction interval crosses zero. Neither recoverability nor lower violation frequency guarantees superior reserve decisions; no transition-localized heterogeneity or allocation advantage is claimed. The bounded deterministic external branch is limited to the evaluated windows and existing weights; it does not close EXP-01 or establish a universal transfer effect. Corrected-v2 is superseded/not reinstated and v4 is invalidated by physical-window mismatch. Finally, MoE functions strictly as an implementation comparator rather than a required mechanism: increasing routing complexity is not the operative source of value under the tested observability regimes. The evaluated evidence supports deterministic rules under clean telemetry, recalibration under observable drift, and consequence information when critical telemetry is withheld, while the current evidence does not establish a transition-specific reserve-sizing advantage. Reserve decisions remain benchmarked against direct observable conditioning. The current evidence therefore supports stopping complexity expansion: state recoverability, risk-shape change, and decision value remain distinct claims. E04 selected no feasible policy; E07 remains inactive.

Acknowledgment

The authors acknowledge the use of OpenAI ChatGPT and Codex for assistance in drafting and refining prose across Sections I–VII, formatting IEEE LaTeX macros, and developing automated verification scripts. All mathematical derivations, experimental designs, empirical results, and scientific interpretations were developed by the authors, who take full responsibility for the contents of this paper.

Code and Data Availability: Raw SCADA datasets (WTB, ENGIE La Haute Borne, Kelmarsh, Penmanshiel) are publicly available. Code, manifests, and verification suites are available at https://github.com/b1ue13e/windfarm-latent-boundary-risk. Archived arrays support bounded accounting and diagnostic checks, but historical weight, complete-input and numerical-environment bindings remain incomplete. The original rented compute instance has been reported destroyed, and its investigation has stopped. File hashes do not certify end-to-end reproduction or deployment safety; the supplementary material and repository reproducibility statement specify the frozen gaps. E07 remains inactive. # References

[1] J. Dowell and P. Pinson, "Very-short-term probabilistic wind power forecasts by sparse vector autoregression," *IEEE Transactions on Smart Grid*, vol. 7, no. 2, pp. 1063–1070, 2016, doi: 10.1109/TSG.2015.2424078.

[2] J. Kruse, E. Cramer, B. Sch"afer, and D. Witthaut, "Physics-informed machine learning for power grid frequency modeling," *PRX Energy*, vol. 2, no. 4, p. 043003, 2023, doi: 10.1103/PRXEnergy.2.043003.

[3] P. Pinson, "Wind energy: Forecasting challenges for its operational management," *Statistical Science*, vol. 28, no. 4, pp. 564–585, 2013, doi: 10.1214/13-sts445.

[4] R. Doherty and M. O'Malley, "A new approach to quantify reserve demand in systems with significant installed wind capacity," *IEEE Transactions on Power Systems*, vol. 20, no. 2, pp. 587–595, 2005, doi: 10.1109/TPWRS.2005.846206.

[5] Y. Wang, F. Zhang, H. Kou, R. Zou, Q. Hu, and J. Wang, "A review of predictive uncertainty modeling techniques and evaluation metrics in probabilistic wind speed and wind power forecasting," *Applied Energy*, vol. 396, p. 126234, 2025, doi: 10.1016/j.apenergy.2025.126234.

[6] J. G. Slootweg, S. W. H. de Haan, H. Polinder, and W. L. Kling, "General model for representing variable speed wind turbines in power system dynamics simulations," *IEEE Transactions on Power Systems*, vol. 18, no. 1, pp. 144–151, 2003, doi: 10.1109/TPWRS.2002.807113.

[7] E. Gaertner *et al.*, "Definition of the IEA wind 15-megawatt offshore reference wind turbine," National Renewable Energy Laboratory (NREL), Golden, CO, NREL/TP-5000-75698, 2020. doi: 10.2172/1603478.

[8] J. Tautz-Weinert and S. J. Watson, "Using SCADA data for wind turbine condition monitoring: A review," *IET Renewable Power Generation*, vol. 11, no. 4, pp. 382–394, 2017, doi: 10.1049/iet-rpg.2016.0248.

[9] M. A. Ullah, K. Mikhaylov, and H. Alves, "Enabling mMTC in remote areas: LoRaWAN and LEO satellite integration for offshore wind farm monitoring," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 6, pp. 3744–3753, 2022, doi: 10.1109/TII.2021.3112386.

[10] S. Daenens, T. Verstraeten, P.-J. Daems, A. Nowé, and J. Helsen, "Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data," *Wind Energy Science*, vol. 10, no. 6, pp. 1137–1152, 2025, doi: 10.5194/wes-10-1137-2025.

[11] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, "Graph WaveNet for deep spatial-temporal graph modeling," in *Proceedings of the 28th international joint conference on artificial intelligence (IJCAI)*, 2019, pp. 1907–1913. doi: 10.24963/ijcai.2019/264.

[12] S. Guo, Y. Lin, N. Feng, C. Song, and H. Wan, "Attention based spatial-temporal graph convolutional networks for traffic flow forecasting," in *Proceedings of the AAAI conference on artificial intelligence*, 2019, pp. 922–929. doi: 10.1609/aaai.v33i01.3301898.

[13] L. Bai, L. Yao, C. Li, X. Wang, and C. Wang, "Adaptive graph convolutional recurrent network for traffic forecasting," in *Advances in neural information processing systems*, 2020, pp. 17804–17815.

[14] J. Park and J. Park, "Physics-induced graph neural network: An application to wind-farm power estimation," *Energy*, vol. 187, p. 115883, 2019, doi: 10.1016/j.energy.2019.115883.

[15] D. Kim, G. Ryu, and C. Moon, "Accuracy of a short-term wind power forecasting model based on deep learning using LiDAR-SCADA integration: A case study of the 400-MW Anholt offshore wind farm," *Applied Energy*, vol. 373, p. 123882, 2024, doi: 10.1016/j.apenergy.2024.123882.

[16] N. Zehtabiyan-Rezaie, A. Iosifidis, and M. Abkar, "Physics-guided machine learning for wind-farm power prediction: Toward interpretability and generalizability," *PRX Energy*, vol. 2, no. 1, p. 013009, 2023, doi: 10.1103/PRXEnergy.2.013009.

[17] E. A. Bossanyi, "The design of closed loop controllers for wind turbines," *Wind Energy*, vol. 3, no. 3, pp. 149–163, 2000, doi: 10.1002/we.34.

[18] F. D. Bianchi, H. De Battista, and R. J. Mantz, *Wind turbine control systems: Principles, modelling and gain scheduling design*. London: Springer, 2006. doi: 10.1007/1-84628-493-7.

[19] L. Y. Pao and K. E. Johnson, "Control of wind turbines: Approaches, challenges, and recent developments," *IEEE Control Systems Magazine*, vol. 31, no. 2, pp. 44–62, 2011, doi: 10.1109/MCS.2010.939962.

[20] C. Zhao *et al.*, "A physics-aware dynamic graph and mixture-of-experts framework for wind power forecasting," *International Journal of Electrical Power & Energy Systems*, vol. 176, p. 111769, 2026, doi: 10.1016/j.ijepes.2026.111769.

[21] J. B. Bremnes, "Probabilistic wind power forecasts using local quantile regression," *Wind Energy*, vol. 7, no. 1, pp. 47–54, 2004, doi: 10.1002/we.107.

[22] H. A. Nielsen, H. Madsen, and T. S. Nielsen, "Using quantile regression to extend an existing wind power forecasting system with probabilistic forecasts," *Wind Energy*, vol. 9, no. 1–2, pp. 95–108, 2006, doi: 10.1002/we.180.

[23] N. Shazeer *et al.*, "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in *International conference on learning representations*, 2017.

[24] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *Journal of Machine Learning Research*, vol. 23, no. 120, pp. 1–39, 2022.

[25] J. Zhou *et al.*, "SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array," *Scientific Data*, vol. 11, p. 649, 2024, doi: 10.1038/s41597-024-03427-5.

[26] H. Hersbach *et al.*, "The ERA5 global reanalysis," *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999–2049, 2020, doi: 10.1002/qj.3803.

[27] B. J. Pierre *et al.*, "Design of the Pacific DC Intertie wide area damping controller," *IEEE Transactions on Power Systems*, vol. 34, no. 5, pp. 3594–3604, 2019, doi: 10.1109/TPWRS.2019.2903782.

[28] G. Ravikumar and M. Govindarasu, "Anomaly detection and mitigation for wide-area damping control using machine learning," *IEEE Transactions on Smart Grid*, vol. 15, no. 6, pp. 5939–5951, 2024, doi: 10.1109/TSG.2020.2995313.